

Event & Concert Booking System

The **Event & Concert Booking System** is a web-based application to facilitate event organizers and users in managing and booking concert tickets. The system should allow administrators to create and manage events, while users can browse, book, and cancel tickets seamlessly. Use code first approach to design database.

2. System Requirements

2.1 Functional Requirements

- **User Authentication:** Admin, Event Organizer, and User roles.
- **Event Management:** Add, update, delete, and view event details.
- **Ticket Booking:** Users can browse events, book, and cancel tickets.

3. Entity Classes & Properties

3.1 Event Entity

- Id
- Name
- Location
- Date
- AvailableSeats
- TicketPrice
- ICollection<TicketBooking> TicketBookings

3.2 User Entity

- Id
- FirstName
- LastName
- Email
- public ICollection<TicketBooking> TicketBookings

3.3 TicketBooking Entity

- Id
- UserId User User
- EventId
- Event Event
- Quantity
- BookingDate
- Status // Confirmed, Canceled

4. Repository Interfaces

4.1 IEventRepository

- Task<IEnumerable<Event>> GetAllEventsAsync();
- Task<Event> GetEventByIdAsync(int id);
- Task AddEventAsync(Event eventEntity);
- Task UpdateEventAsync(Event eventEntity);
- Task DeleteEventAsync(int id);

4.2 ITicketBookingRepository

- Task<IEnumerable<TicketBooking>> GetBookingsByUserIdAsync(int userId);
- Task BookTicketAsync(int userId, int eventId, int quantity);
- Task CancelBookingAsync(int bookingId);

5. Controllers

5.1 EventController

5.2 TicketBookingController

6. Custom Exceptions

6.1 EventNotFoundException

6.2 BookingNotFoundException